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A ^{19}F Solvatomagnetic Comparison Method for the Determination of the α_1 Scale of Solvent Hydrogen-Bond Donation and Its Application to Deep Eutectic Solvents

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Keywords: fluorine NMR spectrometry | hydrogen bonding | solvatochromic parameters | solvent effects

ABSTRACT

We develop a nonsolvatochromic comparison method for the determination of the α_1 scale of solvent hydrogen-bond acidity by means of ^{19}F NMR spectrometry. We compare the ^{19}F chemical shifts of 4-fluoro-2-methylpyridine (as a sensitive hydrogen-bond acceptor probe) and of 4-fluoronitrobenzene (as a chemically similar but less basic reference probe). This so-called ^{19}F solvatomagnetic comparison method yields the hydrogen-bonding contribution to δ (^{19}F)(4-fluoro-2-methylpyridine) that is well correlated to α_1 values obtained from the solvatochromism of Reichardt's betaine dye. Therefore, this solvatomagnetic comparison method is applied to determine the α_1 values of 13 choline chloride–based deep eutectic solvents.

1 | Introduction

Hydrogen-bond donor (HBD) solvents constitute an important class of solvents [1]. Indeed, this category includes often-used classical solvents (e.g., water and alcohols), a number of green solvents (e.g., ethyl lactate and glycerol), many ionic liquids, and all deep eutectic solvents (DESs) [2]. The right choice and safe use of these solvents require the knowledge not only of their physical but also of their chemical properties [3]. Among these, the hydrogen-bond acidity is evidently the most prominent when HBD solvents are considered.

The hydrogen-bond acidity of solvents (generally given the symbol α) has been measured from a variety of HB-sensitive physicochemical properties: the negative solvatochromism of Reichardt's betaine dye B(30) [4], the positive solvatochromism of the phenoxazone dye Nile Red [5, 6], the solvatochromism of an o-*tert*-butylstilbazolium dye [7], the enthalpy of solution of *N*-methylimidazole [8], the ^{13}C solvatomagnetism of pyridine-*N*-oxide [9, 10] or of *N*,*N*-dialkylbenzamides [11], or the ^{31}P solvatomagnetism of trialkylphosphane oxides [12, 13]. Each property has its limitations and advantages. For example, the betaine dye B(30) cannot be used for carboxylic acids because of the loss of its solvatochromic band upon protonation, whereas Nile Red, with a lower pK_a [14], can. Also, the ^{31}P nucleus is more receptive than the ^{13}C one. An important criterion of choice is the possibility to isolate specific solute/HBD solvent hydrogen bond from other solvent effects on the property. Indeed, a simple HBD solvent like 1-octanol can interact with solutes not only in donating hydrogen bonds through the positive hydrogen of its hydroxyl group but also in receiving hydrogen bonds on the lone pairs of its oxygen atom and in interacting by three kinds

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of nonspecific forces: dispersion and induction because of its molecular polarizability and electrostatic forces because of its permanent dipole moment.

The isolation of hydrogen-bond donation effects from other undesired effects has been mainly performed by three methods. The first one is the solvatochromic comparison method [4]. It consists in the comparison of two solvatochromic indicators having similar sensitivity to the effects of non-HBD solvents, one capable, one incapable or less capable of hydrogen bonding. Therefore, a good linear correlation is observed between the wavenumbers of their solvatochromic bands ($\nu_1 = a\ \nu_2 + b$) in a series of varying non-HBD solvents, whereas HBD solvents deviate from the comparison line. These deviations, $\Delta\nu(1-2) = \nu_1 - [a\ \nu_2 + b]$, can be translated into an α scale if one chooses an origin and a format. In the second method, the contribution of nonspecific effects is calculated by applying the linear solvation energy relationship [15] of Equation (1) to the solvent-dependent property *P*:

$$P = P^\circ + s\,\pi^* + d\,\delta + a\,\alpha + b\,\beta \quad (1)$$

wherein π^*, δ, α, and β describe respectively the solvent's dipolarity–polarizability, polarizability correction, HBD ability, and hydrogen-bond acceptor (HBA) ability and s, d, a, and b are the corresponding regression coefficients. The α values are obtained by Equation (2):

$$\alpha = \left[P - (P^\circ + s\,\pi^* + d\,\delta + b\,\beta \right] / a \quad (2)$$

A third method, the difference method [9–11], is used for ^{13}C NMR properties where a difference d_{ij} of the chemical shift δ of carbon *i* with respect to that of carbon *j* ($d_{ij} = \delta_i - \delta_j$) is supposed to depend solely on the HBD ability of solvents. The method has been used for the ^{13}C solvatomagnetism of pyridine-*N*-oxide [9, 10] and of *N*, *N*-dialkylbenzamides [11]. We believe that the comparison method is the most appropriate one because it is based on the fundamental similarity principle. The LSER method depends on the quality of too many parameters and regression coefficients, and the difference method does not always deliver uncontaminated α values [16].

In this work, we choose the ^{19}F chemical shift of 4-fluoro-2-methylpyridine as the HBD solvent-dependent property, and we devise a solvatomagnetic comparison method in order to isolate the HBD effect and determine a pure α_1 (symbol α_1 for Laurence values [16] and α for Kamlet–Taft ones [4]) scale of solvents. Indeed, the ^{19}F nucleus is readily investigated by solution nuclear magnetic resonance (NMR) spectrometry as its receptivity is 4732 greater than ^{13}C. Moreover, Giam and Lyle [17] have observed that the ^{19}F chemical shift of this HBA solute (4-fluoropyridine cannot be studied because of its instability) depends on the hydrogen bonding of HBD solvents to the pyridine nitrogen. The first part is devoted to the search of a fluorobenzene probe with ^{19}F chemical shifts varying similarly to those of 4-fluoro-2-methylpyridine in non-HBD solvents but incapable or less capable to receive hydrogen bonds in HBD solvents and to the use of this pair of probes in a ^{19}F solvatomagnetic comparison method to determine the α_1 values of new solvents. In the second part, we validate these solvatomagnetic values through the comparison with values already obtained by a theoretical PCM-TD-DFT solvatochromic comparison method (free of the shortcomings of the experimental comparison method) [18]. In the last part, we determine the α_1 values of a number of DESs because PCM calculations cannot be applied to this important class of new solvents.

2 | Results and Discussion

2.1 | Choice of the Pair of Probes 4-Fluoro-2-methylpyridine/4-Fluoronitrobenzene

Table 1 lists the solvent-induced ^{19}F chemical shifts of 4-fluoro-2-methylpyridine, fluorobenzene, 4-fluoro-1-(trifluoromethyl)benzene, and 4-fluoronitrobenzene for two categories of solvents: the non-HBD solvents and the HBD solvents. It can be seen that the effect of HBD solvents on the ^{19}F chemical shifts of 4-fluoro-2-methylpyridine is large: ~9 ppm from the weakest HBD acetone to the strongest HBD hexafluoroisopropanol (HFIP). The fluorine atom acts as a sensitive observer of the hydrogen-bonding interaction of HBD solvents with the pyridine nitrogen. However, it does not register only hydrogen-bonding interactions because in the category of non-HBD solvents, a 4 ppm variation is observed from carbon disulfide to tetrahydrofuran. This variation is caused by nonspecific solvent effects and by magnetic effects (bulk magnetic susceptibility of the solvent and anisotropy of the susceptibility of the solvent molecules) [17].

These effects can be subtracted from the global chemical shift of 4-fluoro-2-methylpyridine by the use of a proper probe. Table 2 shows the probes **2**, **3**, and **4** that we have studied. All are similar in size and shape to **1**. Probes **2** and **3** are devoid of HBA ability, their Abraham β_2^H hydrogen-bond basicity parameter [20] being close to zero, whereas **4** keeps an HBA ability through the nitro group ($\beta_2^H = 0.23$) [20, 21], however much weaker than the HBA ability of the pyridine nitrogen atom of **1** ($\beta_2^H = 0.63$) [20, 21]. The ^{19}F chemical shifts of substituted fluorobenzenes are known to depend on the field-inductive and resonance effects of their ring substituent [22], which can be appreciated by their Hammett–Brown substituent constant σ_p^+ [23]. On this criterion, Table 2 shows that the electronic effects of the nitro substituent are closer to those of the aza substituent (for 4-fluoro-2-methylpyridine, the heterocycle is viewed as a substituted benzene nucleus and the heteroatom as the para "aza substituent") than the trifluoromethyl one, so the effects of non-HBD solvents on the ^{19}F chemical shift are expected to be similar in the order **1** ≅ **4** > **3** > > **2**.

If the ^{19}F chemical shift of the primary probe 4-fluoro-2-methylpyridine **1** is plotted against those of the comparison probes **2**, **3**, and **4** for a series of non-HBD solvents, these solvents draw comparison lines of Equations (3)–(5) with determination coefficients r^2, standard deviation *s*, and number of points *n*:

$$-\delta\left(^{19}F\right)(\mathbf{1}) = 0.807(\pm 0.116)\left[-\delta\left(^{19}F\right)(\mathbf{2})\right] - 13.74(\pm 13.09)$$
$$r^2 = 0.828 \qquad s = 0.49\,\text{ppm} \qquad n = 12 \quad (3)$$

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TABLE 1 | Negative of the ^{19}F NMR chemical shifts δ (ppm) of 4-fluoro-2-methylpyridine, fluorobenzene, 4-fluoro-1-(trifluoromethyl)benzene, and 4-fluoronitrobenzene measured in various solvents and given with respect to C_6F_6.

Solvents	$-\delta$ (^{19}F) (4-F-2-MeC_6H_3N) 1	$-\delta$ (^{19}F) (C_6H_5F) 2	$-\delta$ (^{19}F) (4-F$C_6H_4CF_3$) 3	$-\delta$ (^{19}F) (4-F$C_6H_4NO_2$) 4
Non-HBD solvents				
Carbon disulfide	101.84	109.21	104.53	99.67
Tetrachloromethane	103.97	111.48	106.90	102.11
Dimethyl sulfoxide	104.02	112.29		102.02
Benzonitrile	104.22	112.64	107.34	102.21
Chlorobenzene	104.38	112.49	107.42	102.38
Nitrobenzene	104.38	112.99	107.69	102.48
Pyridine	104.49	112.80	107.53	102.48
Anisole	104.78	112.93		103.21
Cyclohexane	105.22	112.21	107.68	103.19
Decaline	105.27		107.29	
Fluorobenzene	105.44		108.25	
Propylene carbonate	105.48	114.08		103.63
Methylcyclohexane	105.51		107.91	
Toluene	105.56		107.94	
1,4-Difluorobenzene	105.78		108.76	
Diethyl ether	106.01	113.52		103.94
Tetrahydrofuran	106.09	113.82	108.67	104.10
HBD solvents				
Acetonitrile	105.68		108.52	
Acetone	105.32	113.52	108.19	103.39
Dichloromethane	104.67	113.77	108.58	102.89
Trichloromethane	103.12	112.93	107.43	101.85
Ethyl lactate	103.46	113.78	108.41	103.54
tert-Amyl alcohol	103.25		107.83	
Methanol	102.91	114.49	109.12	104.15
Cyclohexanol	102.86		107.79	
Ethanol	102.77	113.85	108.55	103.65
2-Propanol	102.47	113.22	108.08	103.00
tert-Butyl alcohol	102.40	112.72		102.48
1-Butanol	102.17	113.11		103.04
Ethylene glycol	101.43	113.71	108.19	103.03
2,2,2-Trifluoroethanol	101.19	115.87	110.88	103.87
Benzyl alcohol	101.05	112.68		102.28
Hexafluoroisopropanol	96.36	117.20	112.24	102.46
n-Butylamine	105.84		108.46	
Piperidine	105.74		108.23	

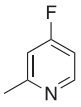
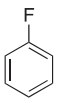
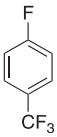
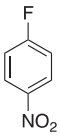
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TABLE 1 | (Continued)

Solvents	$-\delta$ (^{19}F) (4-F-2-MeC$_6$H$_3$N) 1	$-\delta$ (^{19}F) (C$_6$H$_5$F) 2	$-\delta$ (^{19}F) (4-FC$_6$H$_4$CF$_3$) 3	$-\delta$ (^{19}F) (4-FC$_6$H$_4$NO$_2$) 4
N-Methylformamide	104.69		108.25	
N-Methylaniline	103.97		107.76	
Aniline	103.74		107.71	
Formamide	103.01		107.54	
Pyrrole	102.38		107.97	

TABLE 2 | Probes studied for the ^{19}F solvatomagnetic comparison method: 4-fluoro-2-methylpyridine **1**, fluorobenzene **2**, 4-fluoro-1-(trifluoromethyl)benzene **3**, and 4-fluoronitrobenzene **4**. Corresponding hydrogen-bond basicity on the β_2^H scale and substituent constant σ_p^+ of the substituent *para* to the fluorine atom. Determination coefficient r^2 and slope of the comparison line for non-HBD solvents.

				
Probe	**1**	**2**	**3**	**4**
β_2^H	0.63	≅0	≅0	0.23
σ_p^+	1.00[a]	0	0.70[a]	0.83[a]
r^2		0.828	0.924	0.985
Slope		0.807	0.918	0.952

[a]The σ_p^+ values are homogeneously calculated from the carbonyl wavenumber of 4-acetylpyridine, 4-(trifluoromethyl)acetophenone and 4-nitroacetophenone in CCl_4 and the relationship [19]: $\nu(C{=}O) = 1691.3 + 11.1\,\sigma_p^+$ ($n = 17$ substituents, $r^2 = 0.996$).

$$-\delta\left(^{19}\mathrm{F}\right)(\mathbf{1}) = 0.918(\pm 0.080)\left[-\delta\left(^{19}\mathrm{F}\right)(\mathbf{3})\right] + 6.40(\pm 8.55)$$
$$r^2 = 0.924 \qquad s = 0.29\,\mathrm{ppm} \qquad n = 13 \tag{4}$$

$$-\delta\left(^{19}\mathrm{F}\right)(\mathbf{1}) = 0.952(\pm 0.037)\left[-\delta\left(^{19}\mathrm{F}\right)(\mathbf{4})\right] + 6.90(\pm 3.83)$$
$$r^2 = 0.985 \qquad s = 0.15\,\mathrm{ppm} \qquad n = 12 \tag{5}$$

The line of Equation (5) illustrated in Figure 1 comparing the effects of non-HBD solvents on the ^{19}F chemical shifts of 4-fluoro-2-methylpyridine **1** and of 4-fluoronitrobenzene **4** has the highest determination coefficient, the smallest standard deviation, and the slope the closest to unity. Therefore, 4-fluoronitrobenzene **4** is the most similar to 4-fluoro-2-methylpyridine **1** as far as nonhydrogen-bonding effects on the ^{19}F chemical shift is concerned and the pair of probes 4-fluoro-2-methylpyridine **1**/4-fluoronitrobenzene **4** has been chosen for the ^{19}F solvatomagnetic comparison method described below.

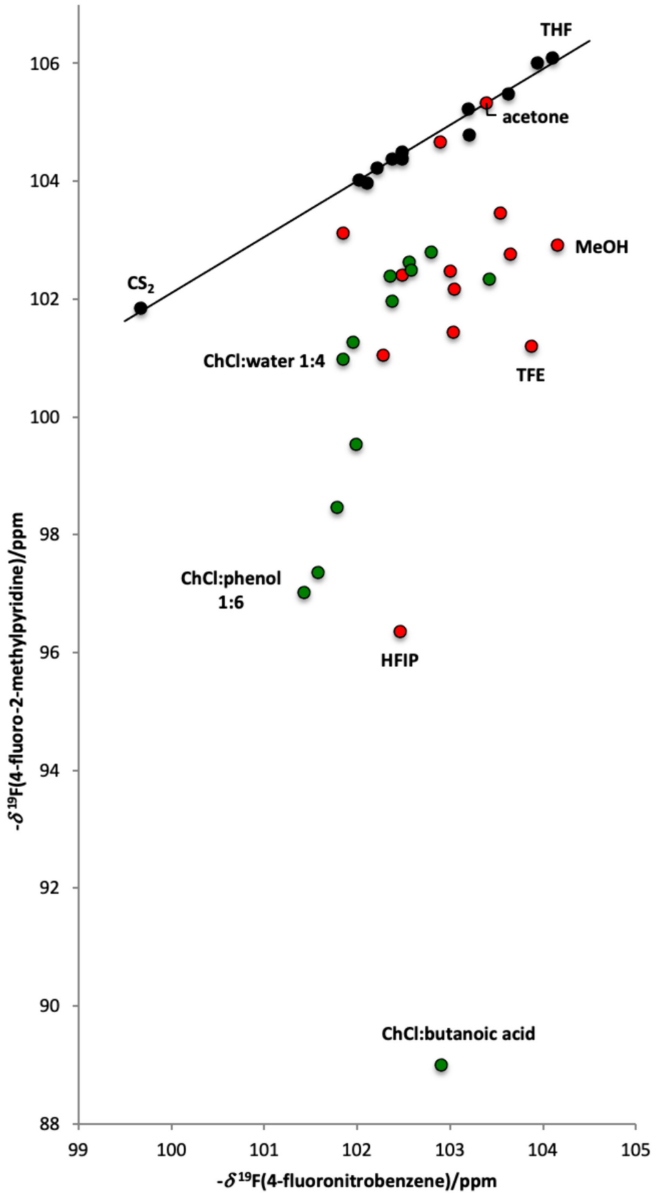


FIGURE 1 | Solvatomagnetic comparison method. Negative of the ^{19}F NMR chemical shift δ of 4-fluoro-2-methylpyridine plotted against negative of the chemical shift δ of 4-fluoronitrobenzene measured in various solvents. Non-HBD solvents fixing the comparison line of Equation (5): black circles (THF, tetrahydrofuran). Conventional HBD solvents: red circles (HFIP, hexafluoroisopropanol; TFE, trifluoroethanol). Choline chloride–based deep eutectic solvents: green circles.

2.2 | The ^{19}F Solvatomagnetic Comparison Method and the α_1 Scale

The addition of data points representing conventional HBD solvents (red circles) shows that they are all, except the very weak

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HBD acetone, displaced from the comparison line in the same direction by amounts depending on the hydrogen-bond donation strength, for example HFIP > TFE > MeOH. These deviations from the comparison line, noted $\Delta\delta$(**1**–**4**), can therefore be interpreted as the contribution to the 4-fluoro-2-methylpyridine ^{19}F chemical shift of the hydrogen bond to the pyridine nitrogen. The results of their calculation by Equation (6) are given in Table 3.

$$\Delta\delta(\mathbf{1}-\mathbf{4}) = \left[-\delta\left(^{19}\mathrm{F}\right)(4-\mathrm{fluoro}-2-\mathrm{methylpyridine})\right] - \left\{0.952\left[-\delta\left(^{19}\mathrm{F}\right)(4-\mathrm{fluoronitrobenzene})\right]+6.90\right\} \quad (6)$$

TABLE 3 | Hydrogen-bonding contribution $\Delta\delta$(**1**–**4**) (ppm) to the ^{19}F chemical shift of 4-fluoro-2-methylpyridine and comparison to solvatochromic Laurence α_1 values [16].

HBD classical solvents	$\Delta\delta$(1–4)	α_1
Acetone	(−0.012)	0.04
Dichloromethane	0.162	0.10
Trichloromethane	0.722	0.20
tert-Butyl alcohol	2.042	0.24
2-Propanol	2.467	0.53
1-Butanol	2.805	0.65
Ethanol	2.785	0.75
Ethyl lactate	1.991	0.78
Benzyl alcohol	3.201	0.79
Methanol	3.121	1.00
Ethane-1,2-diol	3.535	1.05
2,2,2-Trifluoroethanol	4.575	1.36
Hexafluoroisopropanol	8.063	1.86

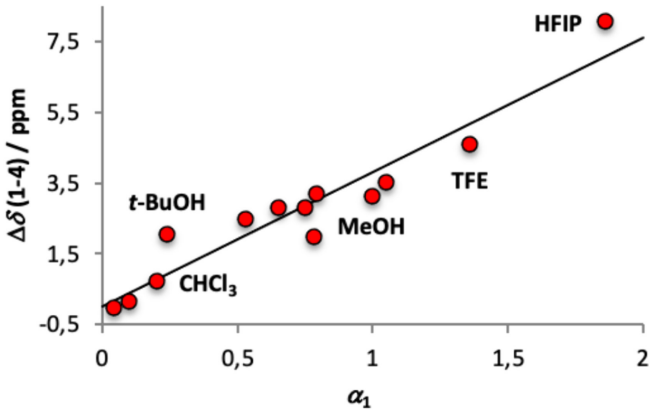


FIGURE 2 | Correlation between the hydrogen-bonding contribution $\Delta\delta$(**1**–**4**) to the ^{19}F chemical shift of 4-fluoro-2-methylpyridine and the theoretical solvatochromic Laurence α_1 scale.

As shown by Figure 2 and Equation (7), these solvatomagnetic $\Delta\delta$(**1**–**4**) defined from the 4-fluoro-2-methylpyridine probe are satisfactorily correlated ($r^2 = 0.910$) to the theoretical solvatochromic Laurence α_1 scale [16, 18] defined from the probe B(30).

$$\Delta\delta(\mathbf{1}-\mathbf{4}) = 3.795(\pm 0.359)\,\alpha_1 - 0.002(\pm 0.315) \quad (7)$$
$$r^2 = 0.910 \qquad s = 0.65\,\mathrm{ppm} \qquad n = 13$$

The determination coefficient rises to 0.939 if the most deviating solvent *tert*-butanol is excluded. We attribute this deviation to a steric effect of the *tert*-butyl group on the hydrogen bonding to the phenolate oxygen of B(30) flanked by two phenyl groups in the ortho positions. In contrast, the nitrogen lone pair of 4-fluoro-2-methylpyridine is hindered only by one methyl group.

Interestingly, the deviations $\Delta\delta$(**1**–**2**) and $\Delta\delta$(**1**–**3**) of HBD solvents from the comparison lines drawn with non-HBD solvents for the comparison of **1** with **2** and of **1** with **3** are less well correlated to the theoretical solvatochromic α_1 values: $r^2 = 0.828$ for $\Delta\delta$(**1**–**2**) vs. α_1 and $r^2 = 0.783$ for $\Delta\delta$(**1**–**3**) vs. α_1. This confirms that 4-fluoronitrobenzene **4** is a better comparison probe than fluorobenzene **2** and 4-fluoro-1-(trifluoromethyl)benzene **3**.

The good correlation between $\Delta\delta$(**1**–**4**) and α_1 shows that the probe 4-fluoro-2-methylpyridine and the ^{19}F solvatomagnetic comparison method deliver an α_1 scale in relative agreement with the probe B(30) and the PCM-TD-DFT theoretical solvatochromic comparison method [18]. Thus, the ^{19}F solvatomagnetic comparison of 4-fluoro-2-methylpyridine and 4-fluoronitrobenzene can be used for the determination of the α_1 scale of neoteric solvents, and in the following, we apply it to DESs for which PCM calculations cannot be performed and in which the zwitterionic dye B(30) is not always soluble or stable.

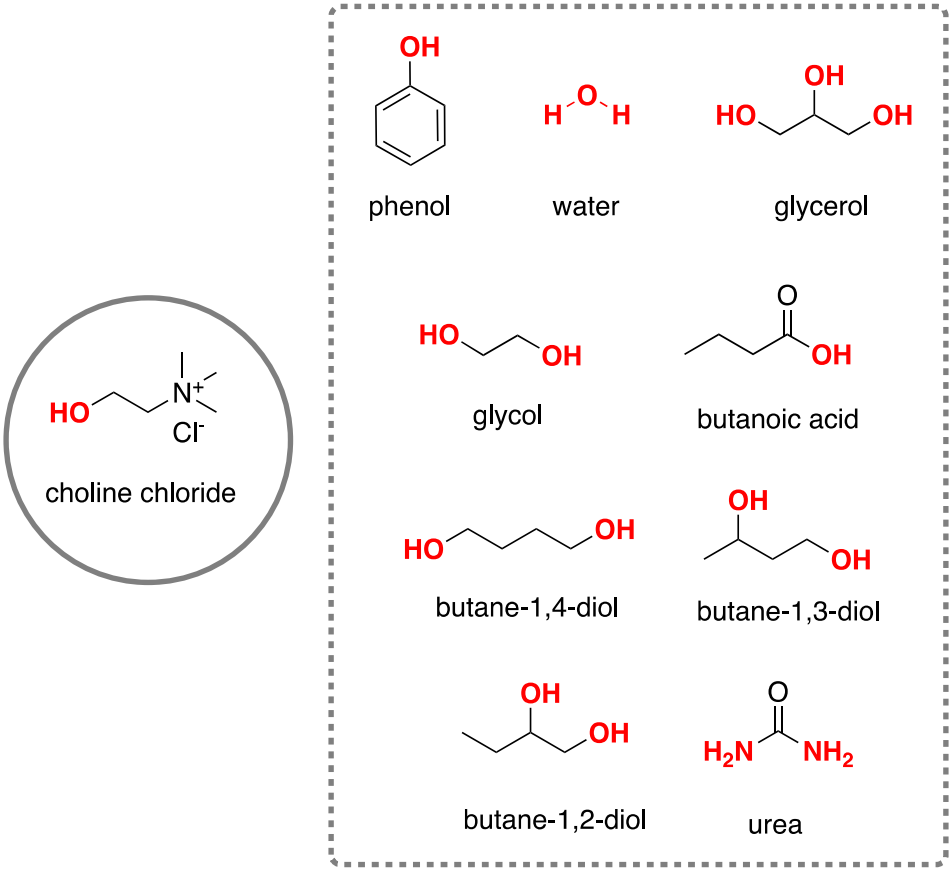
2.3 | Solvatomagnetic α_1 Values of DESs

DESs are liquid eutectic mixtures with low melting points consisted generally of two compounds [2, 24, 25]. They exhibit many "green advantages" over conventional organic solvents and ionic liquids [2, 24, 25]. The DESs being always constituted by an HBD component, the hydrogen-bond donation ability of DESs is among their most important chemical property [26, 27].

We have prepared 13 choline chloride (ChCl)–based deep eutectic solvents [28] by mixing ChCl with HBD solvents (butanediols, glycerol, ethylene glycol, water, phenol, butanoic acid, and urea) in various molar ratios (Scheme 1). The ^{19}F chemical shifts of 4-fluoro-2-methylpyridine and 4-fluoronitrobenzene in these DESs are given in Table 4. The addition of data points (green circles) representing DESs to the graphic of Figure 1 shows that they are displaced from the comparison line in the same direction as HBD conventional solvents, as expected because all these DESs components bring OH or NH_2 groups that are good HBD groups. These deviations $\Delta\delta$(**1**–**4**) calculated by Equation (6) and the corresponding α_1 values calculated by the regression of α_1 to $\Delta\delta$(**1**–**4**) (Equation 8) are given in Table 4.

$$\alpha_1 = 0.240(\pm 0.023)\,\Delta\delta(\mathbf{1}-\mathbf{4}) + 0.065(\pm 0.077) \quad (8)$$
$$r^2 = 0.910 \qquad s = 0.16 \qquad n = 13$$

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SCHEME 1 | Chemical structures of compounds mixed to form the studied deep eutectic solvents (the HBD groups are shown in red).

TABLE 4 | Fluorine NMR chemical shifts δ (^{19}F) (ppm) of 4-fluoro-2-methylpyridine 1 and 4-fluoronitrobenzene 4 with respect to C_6F_6, hydrogen-bonding contribution $\Delta\delta$(**1–4**) (ppm) and solvatomagnetic hydrogen-bond acidity α_1 scale of ChCl-based DESs in order of increasing acidity.

DES (molar ratio)	$-\delta$ (^{19}F) 1	$-\delta$ (^{19}F) 4	$\Delta\delta$(1–4)	α_1
ChCl:butane-1,2-diol (1:4)	102.63	102.56	1.89	0.52
ChCl:glycol (1:2)	102.79	102.80	1.96	0.53
ChCl:butane-1,3-diol (1:4)	102.38	102.35	1.94	0.53
ChCl:butane-1,4-diol (1:4)	102.49	102.58	2.05	0.56
ChCl:glycerol (1:2)	101.97	102.37	2.37	0.63
ChCl:water (1:3)	101.26	101.96	2.69	0.71
ChCl:water (1:4)	100.98	101.85	2.86	0.75
ChCl:urea (1:2)	102.33	103.42	3.01	0.79
ChCl:phenol (1:3)	99.53	101.99	4.45	1.13
ChCl:phenol (1:4)	98.46	101.78	5.32	1.34
ChCl:phenol (1:5)	97.35	101.58	6.24	1.56
ChCl:phenol (1:6)	97.01	101.43	6.43	1.61
ChCl:butanoic acid (1:3)	88.99	102.90	15.85	(3.87)

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TABLE 5 | α values from literature: comparison with solvatomagnetic α_1 values (this work).

DES (molar ratio)	α	α_1
ChCl:glycerol (1:2)	0.937 [29], 0.94 [30], 1.49 [32], 0.93 [35], 0.797 [27], 0.903 [34]	0.63
ChCl:glycol (1:2)	0.88 [30], 1.128 [31], 1.47 [32], 0.89 [35], 0.680 [27], 0.891 [34]	0.53
ChCl:butanoic acid (1:3)	0.8472 [6]	(3.87)
ChCl:urea (1:2)	1.140 [31], 1.42 [32], 0.82 [35], 0.710 [27], 0.860 [34]	0.79
ChCl:butane-1,4-diol (1:4)	0.68 [36]	0.56

TABLE 6 | Hydrogen-bond donation ability α_1 of the HBD components of DESs in order of increasing hydrogen-bond donation, determined by the theoretical or semi-empirical solvatochromic comparison method [16].

DES component	α_1
Butane-1,3-diol	0.80
Butane-1,2-diol	0.83
Butane-1,4-diol	0.85
Urea	0.97[a]
Ethane-1,2-diol, glycol	1.05
Propane-1,2,3-triol, glycerol	1.10
Butanoic acid	1.10[b]
Phenol	1.22[c]
Water	1.54

[a]Value of formamide.
[b]Value from Marcus [3].
[c]Value of 3-methylphenol.

These α_1 values are new for 8 DESs over the 13 studied ones. For the five DESs already determined, the literature α values [6, 29–37] are given in Table 5. Even if often determined from the solvatochromism of the same HBA probe (B30), these values determined by different groups show very large differences: from 0.797 to 1.49 for ChCl:glycerol, 0.68 to 1.47 for ChCl:glycol, and 0.71 to 1.42 for ChCl:urea. Moreover, the literature α values are quite different from the solvatomagnetic α_1 ones (see Table 5): α values are all (except ChCl:butanoic acid, vide infra) very significantly higher than α_1 ones. We have shown in a previous work [16] that for conventional solvents, the literature methods for the determination of α were marred by a number of shortcomings. These shortcomings play also for DESs. So, we recommend the ^{19}F solvatomagnetic comparison method for the determination of the HB acidity of DESs.

A rough understanding of the variation of the α_1 values of studied ChCl-based DESs can be obtained from the comparison with the α_1 values of their HBD components. These values are recalled in Table 6. Urea and phenol being solid at room temperature, their α_1 values cannot be measured, so they are attributed α_1 values of close structure liquids (formamide and 3-methylphenol, respectively). The graphical comparison of Figure 3 shows that mixing the strong HBA choline chloride with the HBDs butane-1,2-diol, glycol, butane-1,3-diol, butane-1,4-diol, glycerol, water, and urea yields α_1 values of the mixture lower than the α_1 values of the pure HBD components. For example, the α_1 values of glycerol (1.10), glycol (1.05), and water (1.54) fall, respectively, to 0.63, 0.51, and 0.71–0.79 (depending on the molar ratio) for the corresponding DESs. This ChCl effect can be attributed to a competition between the chloride anion and the nitrogen atom of the pyridinic probe for the hydrogen-bonding site of the OH or NH_2 groups. A number of these groups may hydrogen bond to the strongly basic anion Cl^- ($\beta_2^H = 1.15$ for NBu_4^+ Cl^-) [38] rather than to the less basic pyridine nitrogen of 4-fluoro-2-methylpyridine ($\beta_2^H = 0.63$). It is noteworthy that, for ChCl:phenol DESs, this competition effect, still effective for the 1:3 molar ratio, disappears for DESs with higher phenol concentration (1:4, 1:5, and 1:6).

The very high α_1 value (3.87) of ChCl:butanoic acid does not measure the HB acidity of this DES. It reveals the protonation of 4-fluoro-2-methylpyridine ($pK_a = 5.27$ in water) [39] by the butanoic acid ($pK_a = 4.82$ in water) [40]. Indeed, protonation of the ring nitrogen withdraws electron density from the ring and causes the fluorine resonance to shift downfield, hence a larger deviation from the reference line and a very high α_1 value. Nevertheless, for the study of solvents with significant proton acidity, 4-fluoro-2-methylpyridine remains a better probe than Reichardt's betaine dye B(30) by virtue of its lower pK_a value (5.27 instead of 8.65 for B(30) [41]). Indeed, we have been able (work in progress) to obtain the solvatomagnetic α_1 value of a number of carboxylic acid–based DESs, for example, $\alpha_1 = 1.04$ for decanoic acid:menthol (1:1) or $\alpha_1 = 1.19$ for octanoic acid:dodecanoic acid (3:1). On the contrary, no RCOOH-based DESs can be measured with B(30) because they protonate the phenolic oxygen and the protonated dye no longer exhibits the solvatochromic absorption band [41].

The literature value [6], $\alpha = 0.8472$, of ChCl:butanoic acid has been obtained from the solvatochromic shift of Nile Red. With a pK_a of 1.00 [14], this dye is less sensitive to protonation than 4-fluoro-2-methypyridine. However, solvatochromic shifts of Nile Red depend strongly on nonspecific effects [6, 27, 36]. Nile Red users [6, 27, 36] measure these nonspecific effects with the Kamlet–Taft polarity–polarizability parameter π^* [42], so the reliability of Nile Red α values depends strongly on the precision of this parameter. Unfortunately, there are

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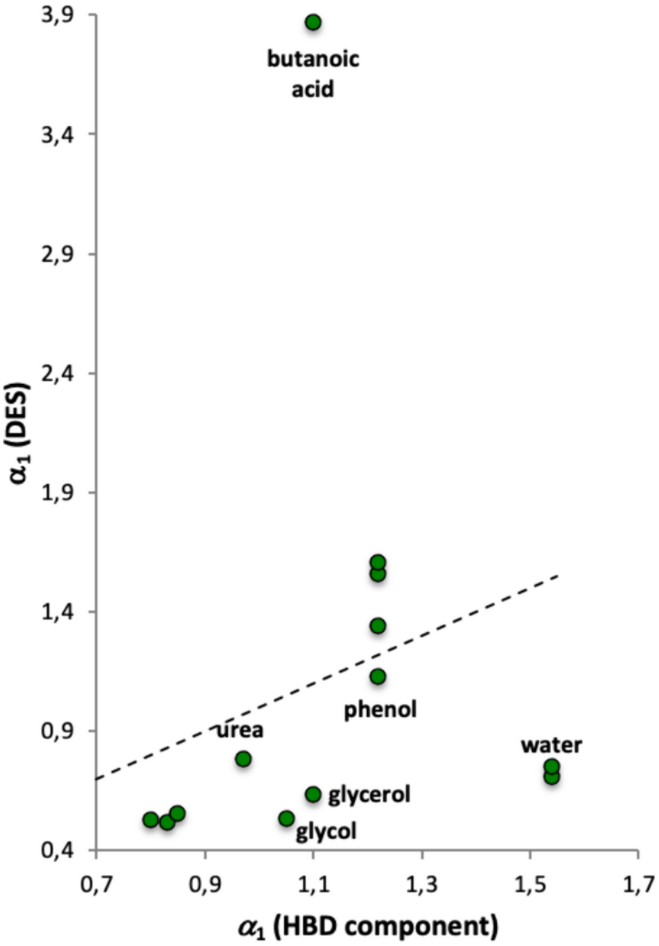


FIGURE 3 | Comparison of the α_1 values of the studied ChCl-based DESs with the α_1 values of their HBD component. The dotted line corresponds to α_1(DES) = α_1(HBD component). Except for ChCl:phenol with high phenol concentration and ChCl:butanoic acid, all DESs fall below the dotted line, showing that the hydrogen-bond acidity of the studied DESs is lower than that of their HBD components.

sometimes significant differences between π^* values for DESs in the literature (e.g., π^* values of 1ChCl:2glycerol range from 0.984 [29] to 1.11 [33]).

3 | Conclusion

Fluorine ^{19}F is readily investigated by solution NMR spectrometry due to its high receptivity. Moreover, in *para*-substituted fluorobenzenes, ^{19}F chemical shifts are quite sensitive to the hydrogen-bonding interaction of the *para* substituent with the solvent. In this work, solute HBA/solvent HBD interactions have been measured by a ^{19}F solvatomagnetic comparison method by means of the pair of probes 4-fluoro-2-methylpyridine/4-fluoro nitrobenzene. This method based on molecular probes yields α_1 values of solvent hydrogen-bond acidity in agreement with those obtained by the theoretical solvatochromic comparison method based on the zwitterionic probe B(30). Its application to choline chloride–based DESs shows that the hydrogen-bond donation ability of the choline chloride:HBD DESs is significantly lowered compared to that of the pure HBD component. As final information, a reviewer has pointed out that the published parameters likely contain nonspecific contributions of volume in addition to the specific HBD ability [43, 44].

4 | Experimental Section

4.1 | Preparation of DESs

The eutectic mixtures were formed by stirring the two components at 100°C in the rigorously measured molar ratios given in Table 4 until a homogeneous liquid was formed (2–4 h).

4.2 | NMR Spectrometry

^{19}F NMR spectra were recorded on a 400-MHz spectrometer. Chemical shifts were referenced to hexafluorobenzene at $\delta = -164.0$ ppm. Solutions of 4-fluoro-2-methylpyridine and 4-fluoronitrobenzene were prepared at concentrations of 0.02–0.07 mol/L depending on solubility. It is possible to prepare only one solution of both 4-fluoro-2-methylpyridine and 4-fluoronitrobenzene as solutes and to record only one NMR

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spectrum of this ternary solution, as the two solutes do not interact at these concentrations.

Author Contributions

Christian Laurence: conceptualization, writing – original draft, methodology. **Sergui Mansour:** investigation. **Daniela Vuluga:** writing – review and editing, supervision. **Julien Legros:** funding acquisition, writing – review and editing, supervision.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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